IS1105

Lithosphere

Booklet 4

INTEGRATED SCIENCES DEPARTMENT
JUNIOR HIGH SCHOOL
NATIONAL JUNIOR COLLEGE

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INTEGRATED SCIENCES DEPARTMENT

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Booklet Contents

By the end of the series of lectures covered within this booklet, students should be able to:

1. Define and explain the processes that create volcanoes (i.e. at convergent and divergent boundaries)
2. Distinguish between the different types of volcanoes formed at the different boundaries
3. Explain the viscosity of lava given its chemical composition
4. Apply concepts of chemical bonding in explaining the physical properties of components found within lava
5. Explain, qualitatively, the composition of different phases of a binary mixture using a phase diagram
6. Explain how the composition of lava differentiates as a function of temperature

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Chapter 4

Volcanology

4.1 Introduction

4.2 Magma

Before we go into volcanoes, it is important to first think about the material that makes up the volcanoes because the differences in magma lead to the formation of different types of volcanoes. The composition of magma varies and indicates the source materials and depth from which they originated.

In general, there are three types of magma, each having different points of origin and characteristic composition: basaltic, andesitic and rhyolitic. The chemical composition, temperature, and the amount of dissolved liquids and gases vary for the three types of magma and each of these characteristics affect the viscosity of magma.

4.2.1 Characteristics

Types of Magma

Types of magma are determined by chemical composition of the magma. Three general types that are recognised are basaltic magma, andesitic magma and rhyolitic magma.

1. Basaltic magma has about 44-55% silica (SiO_2) and is high in iron (Fe), magnesium (Mg) and calcium (Ca) and low in potassium (K) and sodium (Na). Because of its lower silica content, it is basic in nature. Though temperature of magmas is difficult to measure (due to the danger involved), laboratory measurement and limited field observations indicate that the eruption temperature of basaltic magma is between 1000 to 1200°C.

2. Andesitic magma has 55-65% SiO_2 and is intermediate in Fe, Mg, Ca, Na and K. Eruption temperature is about 800 to 1000°C.

3. Rhyolitic magma has SiO_2 65-75% and is low in Fe, Mg and Ca but high in K and Na. Its high silica content makes it acidic in nature. Eruption temperature is the lowest, from about 650 to 800°C.

Gases in Magma

At the depth in the earth, nearly all magmas contain gas dissolved in the liquid. The high pressure deep in the earth keeps the gas in solution in the magma, but when pressure is decreased as magma rises toward the surface of the earth, the gas comes out of the magma and forms a separate gas phase. This is similar to how carbonated beverages are bottled at higher pressure and pressure is decreased when the can is opened and forms a separate gas phase that is seen as bubbles.

Gas gives magma their explosive character, because the volume of gas expands as pressure is reduced. The compositions of gases in magma are mostly water vapour and some carbon dioxide. Minor amounts of sulphur, chlorine and fluorine gases are also found.

The amount of gas in magma is also related to the chemical composition of the magma. Rhyolitic magmas usually have higher gas contents than basaltic magmas, and this is linked to the next property of magma - viscosity.

Viscosity

Viscosity is the resistance to flow and this is the opposite of fluidity. As such, this property determines the ability of the fluid to trap gas within. Viscosity depends primarily on the composition of magma and the temperature. Viscosity increases with increasing SiO_2 concentration in the magma. Viscosity decreases with increasing temperature of the magma.

As such, basaltic magmas tend to be fairly fluid, having low viscosity. Even so, their viscosity is still 10 000 to 100 000 times more viscous than water. Rhyolitic magmas, on the other hand, tend to have even higher viscosity, being between 1 million to 100 million times more viscous than water. Viscosity affects gas content of magma in that the more viscous the magma, the more able it is to trap the gases in it.

4.2.2 How Magmas Form in the Earth

We know that the mantle of the earth is solid. How then are volcanoes able to spew out the flows of lava from the mantle of the earth? There must be some

Magma Type	Solidified Rock	Chemical Composition	Temperature (°C)	Viscosity	Gas Content
Basaltic	Basalt	Basic; 45-55% SiO_2; high in Fe, Mg and Ca; low in K and Na	1000 - 1200	Low	Low
Andesitic	Andesite	55-65% SiO_2; intermediate Fe, Mg, Ca, Na and K	800 - 1000	Intermediate	Intermediate
Rhyolitic	Rhyolite	Acidic; 65-75% SiO_2; low in Fe, Mg and Ca; high in K and Na	650-800	High	High

Table 4.1: Summary table of magma type, chemical composition, temperature and viscosity.

processes that melt the mantle rock, forming the magma that we see oozing out from volcanoes.

Increase in Temperature

This is actually the least important cause of magma formation in the mantle, contrary to popular belief. It must be noted that not all minerals melt at the same temperature. Minerals with high silica content (e.g. quartz) melt at lower temperatures than those with low silica content (e.g. olivine). Since rocks are made up of different minerals, they do not melt at a single temperature but over a range of temperatures.

Decrease in Pressure

Very high pressures in mantle rocks prevent atoms within minerals from breaking chemical bonds and moving freely from one another to form magma. Thus, most rocks within the mantle do not melt even though their temperature may be greater than that necessary to melt the same rocks at the lower pressures of the earths surface.

However, if pressure on mantle rock is decreased, the atoms may move freely from one another, resulting in partial melting of the rock. This process is called pressure-release melting.

At which plate boundary do you think such a process occurs?

Addition of Water

The addition of small amounts of water to the mantle material will result in a decrease in its melting temperature. This is largely due to the electrically polarised nature of a water molecule, which causes a decrease in cation-anion bond strengths within minerals. As such, bonds may be broken at very high temperatures so that atoms may move freely from one another to form magma. This process results in partial melting of the mantle rock.

At which plate boundary do you think such a process occurs?

4.2.3 Origin

Origin of Basaltic Magma

Much evidence suggests that Basaltic magmas result from dry partial melting of mantle.

1. Basalts make up most of oceanic crust and only mantle underlies the crust.
2. Basalts contain minerals like olivine, pyroxene and plagioclase, none of which contain water.
3. Basalts erupt non-explosively, indicating a low gas content and therefore low water content.

The earth's mantle is made of garnet peridotite (a rock made up of olivine, pyroxene and garnet), evidenced by the pieces that are brought to the surface by erupting volcanoes. Laboratory tests are carried out on garnet peridotite to determine its melting behaviour. Under normal conditions, the temperature in the earth is lower than what is required to begin melting the mantle. As such, it is not the temperature of the inside of the earth that causes the mantle to melt. Instead, the mantle undergoes decompression melting, which involves the upward movement of the earth's mantle to an area of lower pressure.

What would cause this upward movement of the earth's mantle?

Since we have established that rocks are made up of many minerals and do not melt at a single temperature, partial melting might take place. The liquids from the partial melt can then be separated from the remaining crystals because liquids have a lower density than solids. Basaltic magmas appear to originate in this way.

Origin of Rhyolitic Magma

Most rhyolitic magma appears to result from wet melting of continental crust. The evidence for this is:

1. Most rhyolites are found in areas of continental crust.
2. When most rhyolitic magma erupts from volcanoes it does so very explosively, indicating high gas content.
3. Solidified rhyolite contains quartz, feldspar, hornblende, biotite, and muscovite. Some of the minerals contain water, indicating high water content.

Still, the temperature in continental crust is usually not high enough to cause the melting of the crust. Thus another heat source is necessary. In most cases it appears that this heat source is basaltic magma. The basaltic magma is generated in the mantle, as discussed above, and rises into the continental crust. But, because basaltic magma has a high density it sometimes stops in the crust and crystallizes, releasing its heat into the surrounding crustal rocks. This may cause wet partial melting of the crust to produce rhyolitic magmas (crustal rocks generally contain water, either in pore spaces or minerals in the rocks).

Origin of Andesitic Magma

Andesitic magmas erupt in areas above subduction zones. This suggests a relationship between the production of andesitic magma and subduction. An earlier theory suggested that wet partial melting of subducted oceanic lithosphere occurs, but newer theories suggest that it is wet partial melting of mantle. Since the oceanic lithosphere is in contact with ocean water, there should be much water in the pore spaces of upper oceanic crustal rocks as well as water contained within clay minerals that have settled to the sea floor. When this material is subducted, it begins to heat up and water is driven off. If the water enters the overlying asthenospheric mantle, it will lower its melting temperatures and thus melting will occur. This melting will produce basaltic magmas with high water content. Basaltic magmas can undergo change as they pass through the earth's crust, particularly if they pass through the thicker and more siliceous continental crust. If the crust gets hot enough, it can melt, as discussed above, and this siliceous melt can mix with the basaltic magma to make an intermediate andesitic magma.

4.3 Physical Features of Volcanoes

4.3.1 Parts of a Volcano

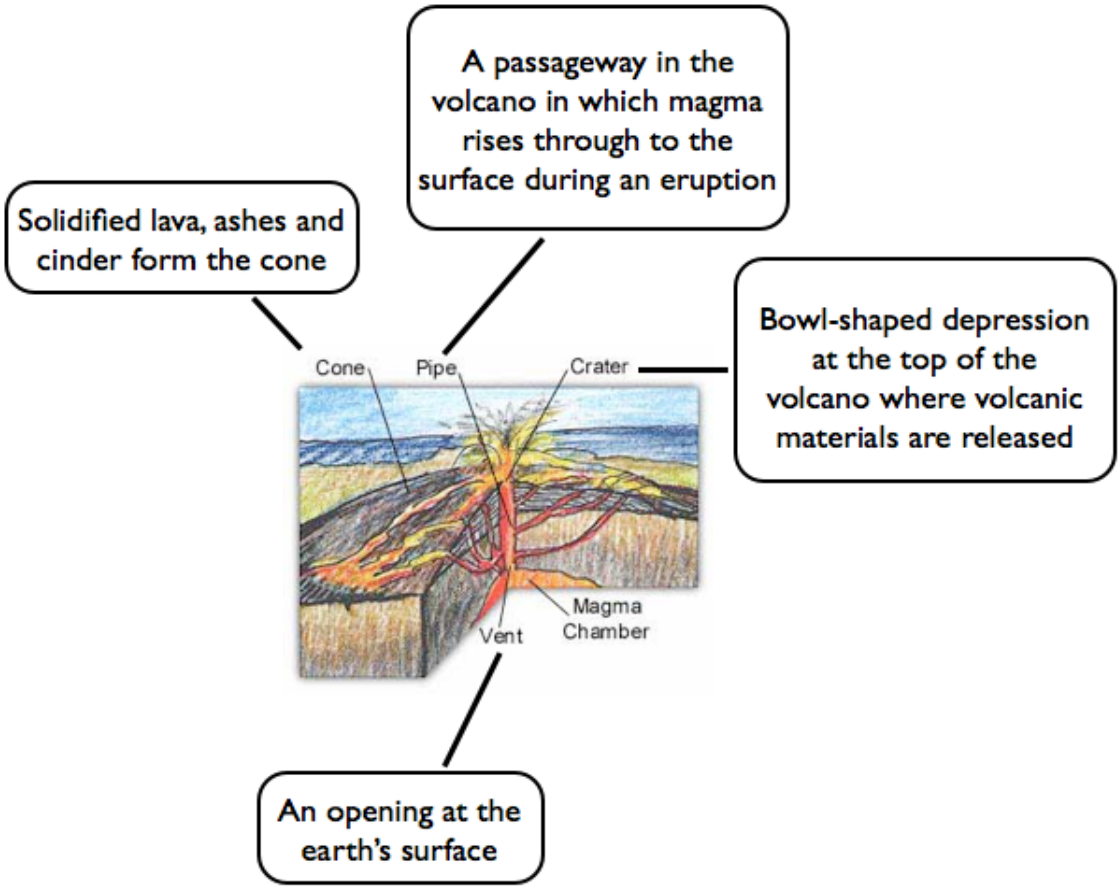


Figure 4.1: Parts of a volcano.

4.3.2 Volcanoes and Plate Tectonics

Get Cracking I - Recap

4.3.3 Types of Volcanoes

Composite Volcanoes

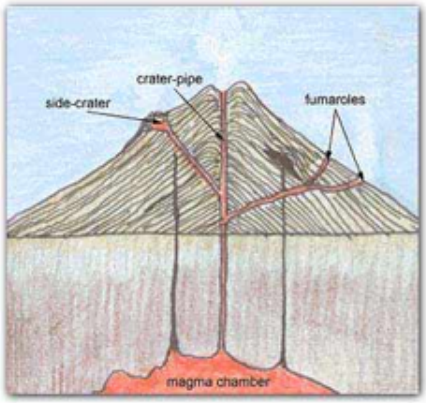


Figure 4.2: Illustration of a composite volcano

Composite volcanoes are also known as strato-volcanoes. They are characterised by alternating layers of lava and rock fragments. This is the reason they are called composite (which indicates it is a combination of two or more materials).

Strato-volcanoes are usually large and conical, often forming impressive, snow-capped peaks which can exceed 2500 m in height and 1000 km^2 in surface area.

These volcanoes usually erupt in an explosive way, which is usually caused by viscous magma. The source of magma of this rock is classified as acidic, or high in silica, to intermediate.

When viscous magma rises to the surface, it usually clogs up the crater-pipe and gas in the crater-pipe gets locked up. Therefore, the pressure will increase, resulting in an explosive eruption.

Shield Volcanoes

Figure 4.3: Illustration of a shield volcano

Shield volcanoes are huge in size. They are built by many layers of runny lava flows. Lava spills out of a central vent or a group of vents. A broad shaped, gently sloping cone is then formed. This is caused by the very fluid basaltic lava which cannot be piled up into steep mounds.

At which plate boundary do you think you would find such volcanoes?

Shield volcanoes may also be produced by hot spots which lie far away from the edges of tectonic plates.

The eruptions of shield volcanoes are characterised by low-explosivity lava-fountaining that forms cinder cones and spatter cones at the vent. Famous shield volcanoes include that of Mauna Loa and Kilauea in Hawaii.

Cinder Cone

A cinder cone is a steep conical hill formed above a vent. Cinder cones are among the most common volcanic landforms formed in the world. However, they are less destructive and usually do not cause any loss of life. The cones usually form in groups and often occur on the flanks of strato-volcanoes and shield volcanoes.

Cinder cones are built from lava fragments called cinders. The lava fragments are ejected from a single vent and accumulate around the vent when they fall back to earth.

Cinder cones grow rapidly and soon approach their maximum size. They rarely exceed 250 m in height and 500 m in diameter.

Figure 4.4: Illustration of a cinder cone

A great example of a cinder cone is Paricutin in Mexico. It was born in February 20, 1943 in a corn field and grew to 300 feet in 5 days.

Spatter Cone

When hot erupting lava contains just enough explosive gas to prevent the formation of a lava flow, but not enough to shatter it into small fragments, the lava is

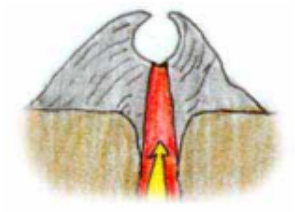


Figure 4.5: Illustration of a spatter cone

torn by expanding gases into fluid hot clots, ranging in size from 1 cm to 50 cm across, called spatter.

When spatter falls back to earth, the clots weld themselves together and solidify, forming steep-sided accumulations. These accumulations focused on an individual vent are called spatter cones.

4.4 Compositional Mayhem

4.4.1 Phase Diagrams

In trying to understand the different phases (e.g. solid, liquid and gas) that are inherent within the earth, and how the dominance of these phases vary as a function of temperature, pressure and composition, we should first begin by analysing a simple binary mixture.

You already know that a mixture refers to a system where two or more substances are physically (and not chemically) combined together. Again, we should try and understand what this really means. How can we define a physical interaction and in the context of this definition of a physical interaction, what is, or how can we, classify chemical interactions? To start thinking about this question, we must first ask ourselves what an interaction represents? Interactions refer to attractions and repulsions (much like social interactions, or a host of other forms of interactions that you might think of). And what is an attraction or repulsion really? Energy. In a system where attraction is dominant, it takes some effort or work to remove that attraction; in a system where repulsion is dominant, the system itself is trying to remove that repulsion. Attraction results in a lowering of energy, repulsion results in a raising of energy.

Therefore, the notion of energy itself would allow us to start thinking about mixtures and how mixtures change as a function of temperature, pressure and composition since these factors would affect the energy of the system itself. This notion of energy implies that there is always some form of *balance* or *equilibrium*; the extent to which a change occurs would invariably depend on the temperature, pressure, and composition of the system of interest.

We begin our discussion by considering a simple mixture of two substances. We call this a binary (referring to two substances) mixture. Let's not be too fancy for now. Let's simply consider A mixed with B. And again, let's just consider how the variation in temperature and composition affect what we would find. We need not complicate our discussion here by considering the pressure dimension, but that can be done quite easily as well.

Figure 4.6 shows the temperature-composition phase diagram for two almost immiscible solids and their completely miscible liquids. Just a bunch of interesting terminology. But what is the idea here, and more importantly, what story can we tell from just looking at such diagrams?

There are four regions that have been labeled in Figure 4.6: a liquid mixture phase, a liquid mixture and solid A phase, a liquid mixture and solid B phase, and a solid mixture phase. There's more beauty in this seemingly simple diagram. It

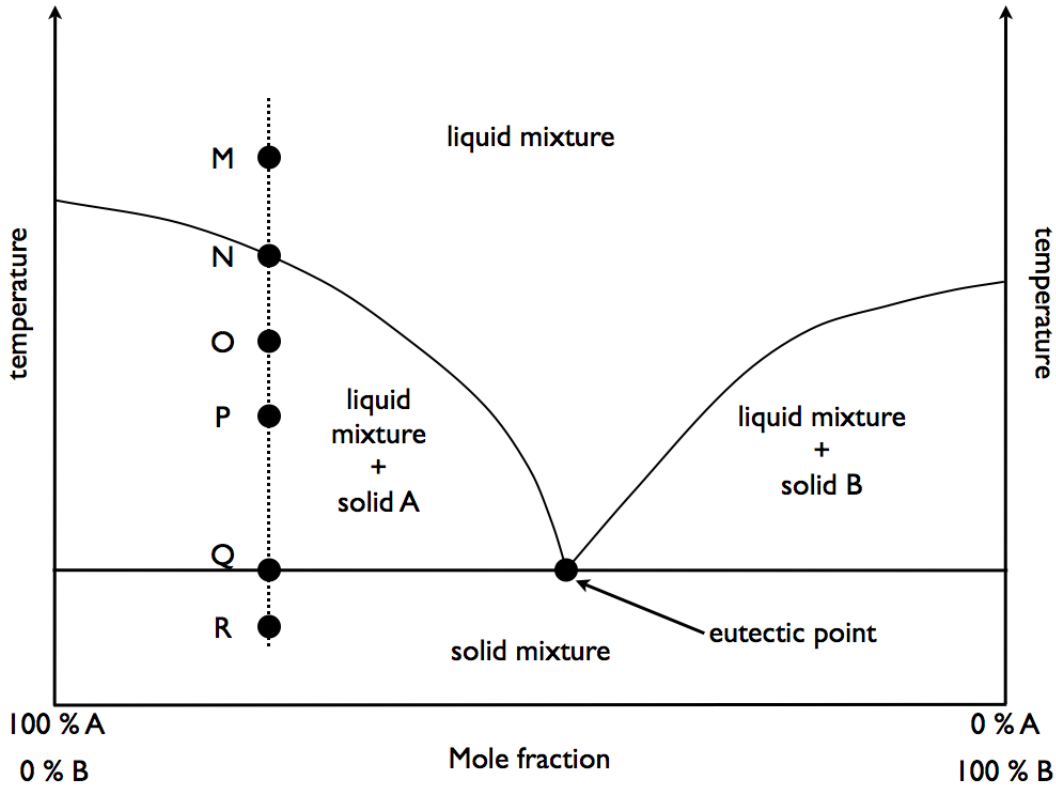


Figure 4.6: Phase diagram of a binary mixture

actually tells us a bunch of stories. Let's just look at one of these stories.

We will use the dotted line, and the points M through to R for quick reference to tell our little story here. You should note that the labels M through to R have no particular significance; I did not start with point A to avoid confusion between substance A and say point A. So say we start at point M. What's in there? (You really should be asking such questions whenever you approach any problem.) At M, we are at some temperature (we will ignore pressure for now) where we have a liquid mixture of both A and B. And how much of each do we have? Say we have mixture where the *mole fraction* of A is 75% and the *mole fraction* of B is 25%. What does this mean? Well, it just means that for every B you find, you have 3 of A. It's as simple as that.

We begin to cool this liquid mixture. As it cools, the two substances will experience different (whether this difference is substantial or not depends on the difference in the individual properties of A and B of course) cooling effects. At point N, something interesting occurs. At this composition, it is now sufficiently 'cold' that some of A would precipitate out of the liquid mixture, i.e. we begin to find some solid A. You must note that this *does not* imply that all of A has become a solid, leaving behind just liquid B. Only some of A begins to precipitate out of

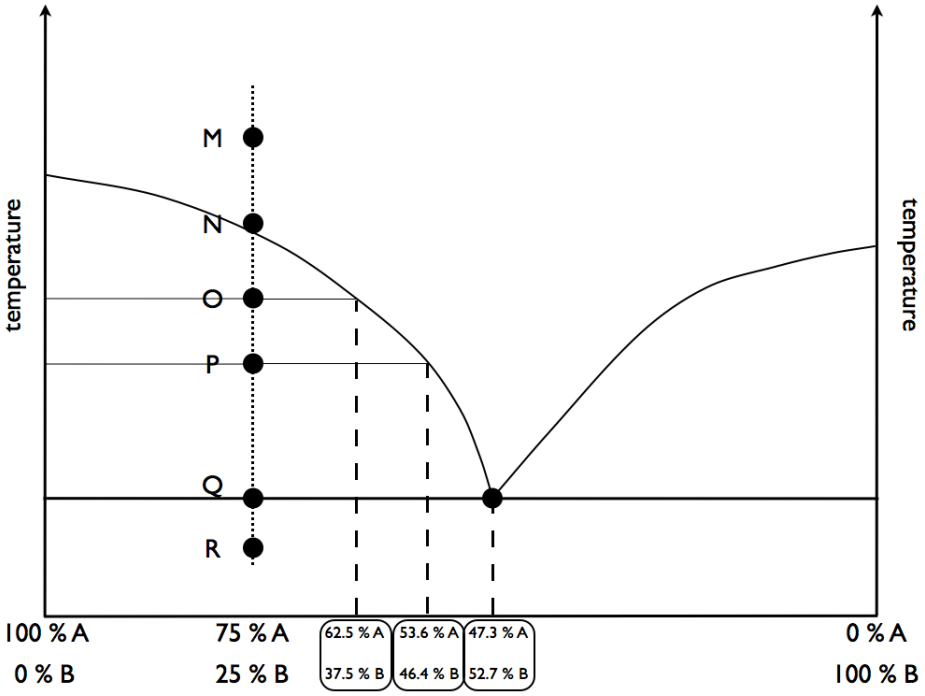
solution at this temperature. Let's trudge on down this dotted line.

We cool this system even more and we get to point O. Now points O and P are all within the area of this diagram. Any point within this region of the diagram would imply that you have some solid A and a liquid mixture of both A and B. Again the question is what is in the solid and what is in this liquid mixture? And we might even ask how much of solid and liquid do we now have? Well, the former is quite a trivial question: we only have solid A. You should note that this situation applies for this particular scenario; you should extend what you learn here to appropriate situations to think about what might be present in other types of phase diagrams of course (which can get pretty interesting). So we have a solid that is 100% A. What about the liquid mixture? It would make sense that since some A has been precipitated out of the mixture that there would be a lower fraction of A left in this liquid mixture. Therefore, the mole fraction of A in the liquid mixture that remains would be less compared to what we had initially, and the mole fraction of B in the liquid mixture would be more compared to what we had initially. This again is trivial. What is not so trivial is the question of the relative quantities of solid and liquid present. We will return to answering the question of the composition of the liquid and the relative quantities of the solid and liquid in the next Get Cracking Exercise. Let's discuss point P first.

We cool the system even more and get to point P. The concepts involved in understanding what point P represents is no different from that of point O. This is because both O and P are within the same region of this phase diagram. The difference is that we are now at a lower temperature, i.e. the system is cooler compared to point P, and we would therefore expect that more of A would precipitate out of the liquid mixture. Thus we have more solid A, and the liquid mixture left would have an even lower fraction of A compared to points M, N and O; likewise, there would be a higher fraction of B compared to points M, N and O.

Let's think a little about how we can figure out the compositions of the liquid mixture associated with points O and P.

Get Cracking II - The use of levers in phase diagrams



Horizontal or 'lever' lines are drawn for the purpose of figuring out the relative quantities of two phases. They also help in telling us something about the composition inherent within each phase. Therefore, within the space labelled 'liquid mixture + solid A' we find that we have some solid A and a liquid mixture that has a greater fraction of B (and therefore a lower fraction of A) compared to the original mixture we began with.

Let's return to our story. We cool the system even more and get to point Q. Point Q is perhaps even more interesting. Here, we hit another equilibrium line. If we draw horizontal lines once again to the so-called 'equilibrium lines' we find that we do have solid A, but something rather interesting occurs at the other end of this horizontal: we hit a point of intersection of three lines. What do we do? This point represents a unique composition of this mixture: we refer to this as the *eutectic composition* and for obvious reasons, this is called the eutectic point. What is a *eutectic point*? Imagine another dotted vertical line drawn through the eutectic point. What's so special about this point? It should be significant enough that we decide to give it a name? Think about this for a moment. At this particular *eutectic composition*, what is different compared to all other compositions? We can make use of our earlier example to think about this. You would find that any other composition, cooling would give rise to some solid precipitating out of the liquid mixture leaving behind a liquid mixture that is enriched in the other component. You would always have some liquid mixture left behind. However, at the eutectic composition, you have a single point, that we call the *eutectic point*, at which all of the liquid mixture turns into a solid mixture. It is almost *as though* the material behaves as a pure substance even though we know it is not: it freezes at a single temperature and *not* over a range of temperatures.

And what happens when we cool this system even more? We get to point R. Within this region of the phase diagram, all we have is an immiscible mixture of solid A and B. Does the amount of each solid change with temperature? Well, if you draw a couple of horizontal lines from a point like R within this region of the phase diagram, you would find that the relative lengths of the horizontals from the point to either vertical axis remains the same. In other words, the relative amounts of solids A and B do not change: we would still have 3 parts of A to 1 part of B.

4.4.2 Understanding lava

So what's the big deal of all of this in the greater scheme of understanding lava and volcanoes? If we were to simplify and confine our arguments here to a variation in temperature, which we know occurs within the earth (based on a whole bunch of data collected over an extended period of time), and the fact that there is

bound to be a variation in the composition of material at different locations, then we can understand why some lava has more liquid silica content. At particular temperatures (and pressures), the silica would precipitate out of the liquid mixture and not flow. The same goes for other minerals and substances that are inherent within the earth. This results in a variation in silica content. You should note that this is of course an oversimplification of what occurs in the earth itself. The concepts are fundamental. However, the number of variables that extend to deep within the earth would make analysis rather difficult. The key point you should take away from this is that you are now able to explain how mixtures change in composition when they freeze or melt. You could even extend this to understanding how fractional distillation works, and you really should think about this. The *ideas* are the same; they are just applied in a different way.

4.4.3 The Importance of Silica Content

Silica, SiO_2 has a giant covalent structure. We have already understood the arrangement of the Si and O atoms in the particular case of sand: there is structure and pattern. Now when we heat this to a sufficient high temperature and lower the pressure, the SiO_2 loses structure and becomes *amorphous*. Nonetheless, there is still an extensive network of covalent bonds. What does this imply and why should we even care about this? Think about this a little. What would it mean to still have an extensive network of covalent bonds? And how does this apply in the particular context of thinking about material flow?

Get Cracking III - Crystalline and amorphous silica

4.5 Piecing everything together

What does the science imply? Or perhaps we should ask, "How does our understanding of all things micro help explain the macroscopic phenomena we can observe?" We first started our discussion by investigating what happens when a mixture of substances cool. In the context of the stuff found within the earth, we can sort of understand how various substances are precipitated out at varying degrees at various temperatures. It turns out that at certain points, a lot more of 'stuff' other than silica precipitates out thus leaving behind lava that is rich in silica; likewise, at other points, silica precipitates out a lot more thus leaving behind lava that is poor in silica content. It is this process of *fractional cyrstalisation* that gives rise to magma with varying content of material at different points within the earth's structure. And what is the significance here? Why do we keep pondering silica content?

Molten silica is an extended material. It is still considered to have a giant covalent structure. We know from chemical bonding that this implies an extensive network of covalent bonds. Now why is this important? This extensive network of strong covalent bonds imply that silicon and oxygen atoms are bonded strongly to one another and in turn implies that lava rich in silica would flow slowly. In other words, the higher the content of silica, the more viscous the lava. You should note that in some texts, the author might suggest the following statement: "The more acidic the lava, the more viscous it is". This statement is not wrong. But you need to understand what acidic means. At present, you might identify acidity with the ability of a particular substance to donate protons (i.e. H^+). This is one definition of acidity. Another notion of acidity is the Lewis definition of an acid: an acid is any substance that is able to accept an electron pair. So let's look at silica, SiO_2. It seems completely saturated with bonds, and there should be no reason for it to accept a lone pair of electrons. So why is it acidic, even under the schema proposed by Lewis? Let's look at silicon. Silicon is in the third period of the periodic table. That implies that it has 3s, 3p and 3d orbitals. And we know that it's 3d orbitals are not used at present. That however *does not mean that they are not there*; they are just simply not utilised. This implies that the silicon atom in silicon dioxide can accept electron pairs, thus we can conclude that silica is acidic. And so we have it: lava rich in silica is more acidic and we know that lava that is rich in silica is also highly viscous owing to the fact that there is an extensive network of covalent bonds which render flow of lava slow.

4.6 Tutorial

1. Explain why the shape of stratovolcanoes is different from that of shield volcanoes.

2. Study the location of Colima Volcano shown in Figure 4.7. Using your

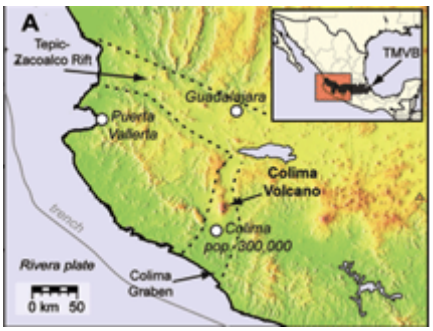


Figure 4.7: Location of Colima Volcano

understanding of the Plate Tectonics Theory, state and explain clearly,

   (a) the type of volcano that Colima Volcano is, and

   (b) the type of magma it produces

3. Hotspots are volcanic features that do not form near tectonic plate boundaries as what we are familiar with but rather, form within plates. This is from a fixed spot where upwelling of magma from deep within the mantle. An example is that of he Hawaiian Islands, as seen in Figure 4.8.

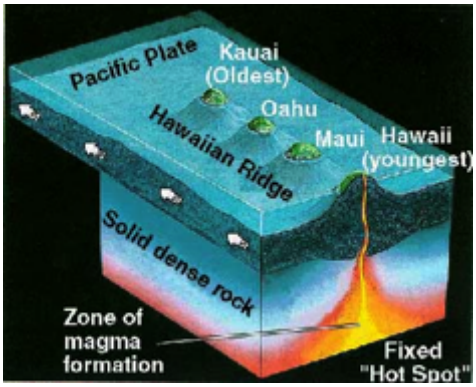


Figure 4.8: Formation of Hawaiian Islands

   (a) What kind of magma do you think forms at hotspot (note that hotspots occur both under oceanic plates as well as continental plates)?

   (b) Explain your answer.